

1.1 Understanding Computer Hardware CPU, GPU, TPU, Memory & DDR

Edge AI Engineering Bootcamp | Analog Data

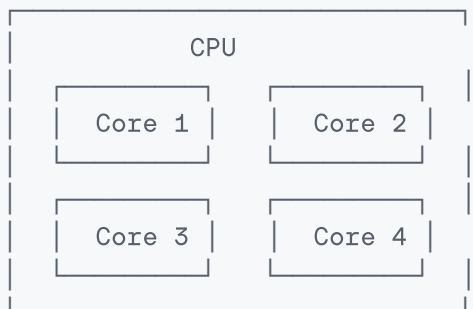
Module 1.1 Companion Notes — Raspberry Pi Ecosystem

Why This Matters

Before we compare Raspberry Pi boards, we need to understand the actual parts inside a computer. Every computer — a laptop, a phone, or a tiny Raspberry Pi — is built from the same basic building blocks. Once you understand these, comparing hardware becomes easy instead of confusing.

1. CPU — Central Processing Unit

The **CPU** is the main "thinking" part of the computer. It reads instructions one step at a time and carries them out. It can do maths, make decisions, and control every other part of the system.



Example: A Raspberry Pi 4B has a **quad-core Cortex-A72 CPU**. "Quad-core" means it has 4 separate processing units inside it, so it can work on 4 things at once instead of just 1.

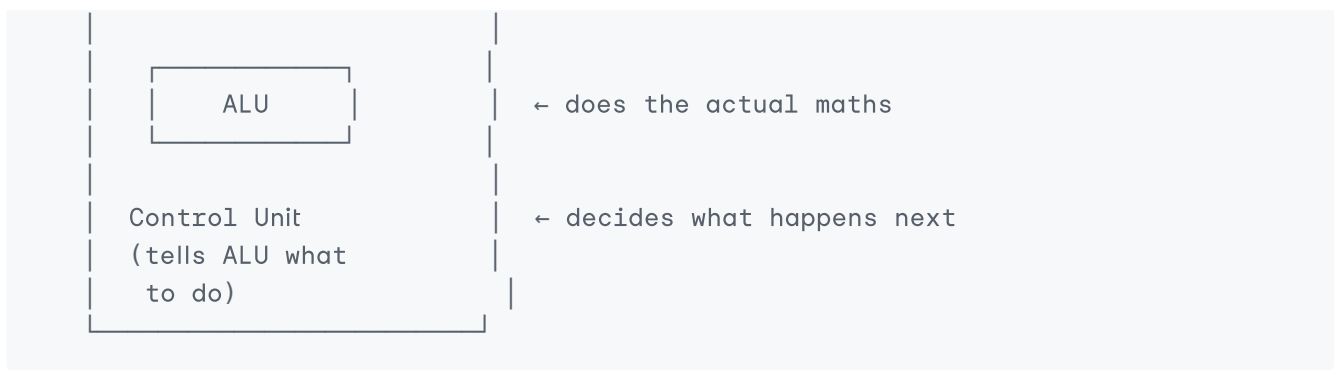
- More cores = more tasks handled at the same time
- Higher clock speed (measured in GHz) = each core works faster

Real example: If you're running a program that plays music AND checks a sensor AND updates a screen, having 4 cores means each task can run on its own core instead of all three fighting over 1 core.

2. ALU — Arithmetic Logic Unit

The **ALU** lives *inside* the CPU. It's the part that actually does the maths and comparisons.

The diagram shows a light blue rectangular area representing the CPU. Inside this area, there is a horizontal line with short vertical bars at each end, representing the ALU. The label "CPU" is positioned above this line.



Every time the CPU does something like:

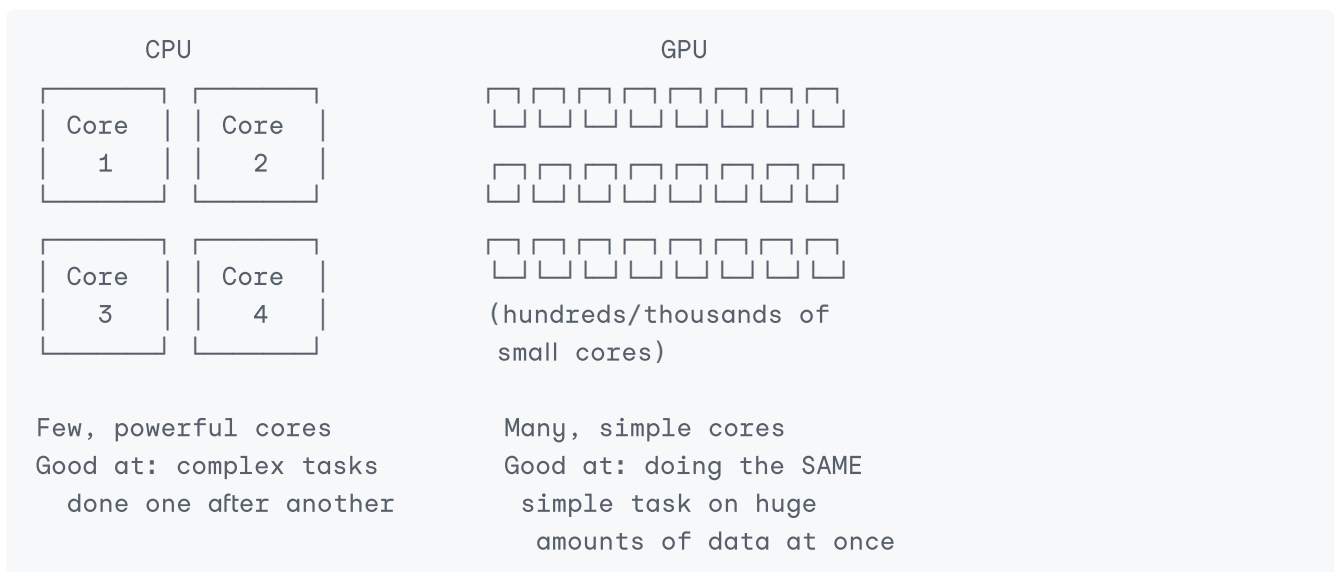
- $5 + 3$
- Is x greater than y ?
- 10×4

...it's the ALU doing that specific calculation. The rest of the CPU (called the **Control Unit**) just decides *what* instruction to send to the ALU next.

Simple way to remember it: CPU = the whole worker. ALU = just the part of the worker that does maths.

3. GPU — Graphics Processing Unit

A **GPU** is built completely differently from a CPU. Instead of a few very powerful cores, it has **thousands of small, simple cores** that all work at the same time.



Example: A screen has millions of pixels. Setting the color of each pixel is a *simple* task, but there are millions of them to do at once. A GPU can hand each pixel to a different tiny core and finish the whole screen almost instantly. This is why GPUs are great for video, graphics, and games.

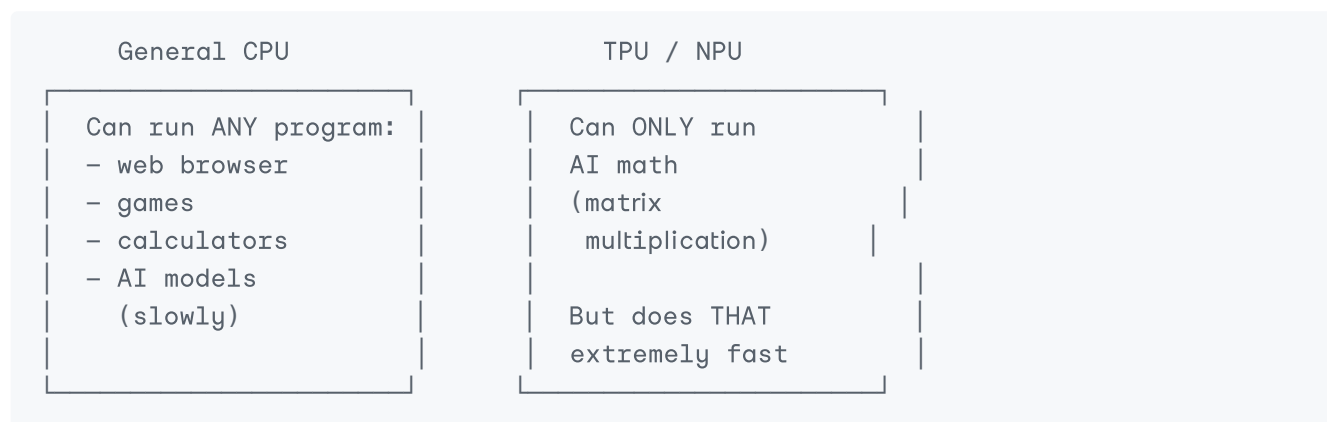
On a Raspberry Pi, the GPU is called **VideoCore** (VideoCore VI on Pi 4B, VideoCore VII on Pi 5). It's mainly used for:

- Decoding/playing video (H.264, H.265)
- Rendering graphics on a display

Important: A Raspberry Pi's GPU does **not** have proper support for running AI models efficiently. Unlike a Jetson (which uses CUDA cores) or a dedicated AI chip, VideoCore was built for video and graphics — not for machine learning maths.

4. TPU / NPU — Tensor Processing Unit / Neural Processing Unit

A **TPU** (or **NPU**) is a chip built for *one job only*: running the specific type of maths used in AI models (matrix multiplication, mostly).



Example: A **Hailo-8** chip (which you can attach to a Raspberry Pi 5 via its PCIe slot) is a good example of an NPU. It can do around 26 TOPS (Tera Operations Per Second) of AI-specific math — far more than a Pi's CPU alone could ever manage — but it can *only* do AI inference. It can't run your operating system or a web browser.

Why this matters for Edge AI: A bare Raspberry Pi has no TPU/NPU. All AI inference runs on the CPU. Adding a Hailo or Coral accelerator gives the board a dedicated AI "engine," which is much faster and more power-efficient for that one job.

5. ROM — Read-Only Memory

ROM stores information that doesn't change and stays saved even when the power is off.



Example: The very first instructions a computer follows the moment it's switched on (before it even loads an operating system) are usually stored in ROM or ROM-like memory. On a Raspberry Pi, this is the **bootloader firmware**.

6. RAM — Random Access Memory

RAM is where the computer keeps information it's using *right now* — the current program, the data it's working on, temporary calculations.

RAM

"Working memory"

- fast

- TEMPORARY

- erased when power turns off

"Random Access" means: the CPU can jump straight to any piece of data in RAM instantly — it doesn't have to search through everything in order, the way you'd flip through pages in a book. It can go directly to exactly the piece of information it needs.

Example: If you open 3 apps on a phone, all 3 apps' current data sits in RAM. Close the apps or turn off the phone, and that RAM data is gone — but the apps themselves (stored in storage/ROM-like memory) are still there, ready to open again.

More RAM = you can keep more things "in progress" at once, and access them faster than if the CPU had to fetch everything from slower storage each time.

7. Types of RAM

Not all RAM is built the same way. Here are the main types you should know:

Type	Full Name	Key Trait	Where You'll Find It
DRAM	Dynamic RAM	Needs to be constantly "refreshed" (re-written) thousands of times per second or it loses data	Main system memory in most computers
SRAM	Static RAM	Faster than DRAM, doesn't need refreshing, but more expensive and takes more space	CPU cache (small, super-fast memory built into the CPU itself)
LPDDR	Low Power DDR	A version of DDR RAM designed to use very little power	Phones, tablets, Raspberry Pi (battery/power-conscious devices)

Speed:

SRAM > DRAM/LPDDR

Size:

SRAM < DRAM/LPDDR

(SRAM is small but very fast; DRAM is bigger but slower)

Example: Your Raspberry Pi 4B uses **LPDDR4**. The Raspberry Pi 5 uses the newer, faster **LPDDR4X** — same basic technology, but improved for higher speed and lower power draw.

8. DDR — Double Data Rate (Explained Properly)

This is one of the most important — and most misunderstood — concepts. Let's break it down step by step.

The old way: SDR (Single Data Rate)

Every piece of memory works on a "clock" — a signal that ticks like a metronome. Old-style memory (SDR) could only send **one piece of data per full clock cycle**.

Clock signal: 

↑ ↑ ↑ ↑
SDR sends data: [D1] [D2] [D3] [D4]

→ 1 piece of data per full tick

The new way: DDR (Double Data Rate)

DDR memory is smarter. Instead of sending data only once per clock cycle, it sends data on **both the rising edge AND the falling edge** of each clock tick.

Clock signal: 

↑ ↓ ↑ ↓ ↑ ↓ ↑ ↓
DDR sends data: [D1] [D2] [D3] [D4] [D5] [D6] [D7] [D8]

→ 2 pieces of data per full tick (DOUBLE the data,
using the SAME clock speed)

In plain terms: Without making the clock run any faster, DDR memory transfers **twice as much data** in the same amount of time, simply by being clever about *when* it sends data — at both the "up" moment and the "down" moment of each clock tick, instead of just once.

Why the numbers keep growing (DDR2, DDR3, DDR4...)

Each new generation of DDR memory (DDR2 → DDR3 → DDR4 → DDR5) finds ways to move even more data per second — through higher clock speeds, wider data paths, and other improvements — while often using *less* power than the generation before it.

Generation	Roughly When Introduced	Key Improvement
DDR2	~2003	Doubled DDR speed, lower power
DDR3	~2007	Higher speeds, lower voltage
DDR4	~2014	Even higher speeds, better efficiency
LPDDR4	Mobile-optimized DDR4	Very low power — used in Pi 4B
LPDDR4X	Improved LPDDR4	Even lower power, higher speed — used in Pi 5

9. How It All Fits Together — One Example

Let's say a Raspberry Pi is running a program that identifies whether a photo shows a cat or a dog.

1. Power on

↳ ROM/bootloader gives the first startup instructions

2. Program + photo data loaded

↳ Stored in RAM (LPDDR4/LPDDR4X) for fast access

3. CPU processes the instructions

↳ ALU inside the CPU does the actual number crunching
(helped by a speed trick called NEON – lets the CPU
do several calculations in a single step)

4. If a display is involved

↳ GPU (VideoCore) renders/updates what's shown on screen

5. If an AI accelerator (like a Hailo chip) is attached

↳ TPU/NPU handles the AI-specific maths much faster
than the CPU alone could

6. Speed of ALL of this

↳ Limited partly by how fast RAM can deliver data to
the CPU – this is why memory bandwidth matters so much

Quick Recap Table

Term	What It Does	Simple Analogy
CPU	General-purpose processor, runs instructions	The main worker who can do many kinds of jobs
ALU	Part of the CPU that does actual maths	The worker's hands doing the calculating
GPU	Many simple cores, great for parallel tasks (graphics/video)	A large team doing the same simple task on thousands of items at once
TPU/NPU	Specialized chip, only does AI math, extremely fast at it	A specialist who does one job better than anyone else, but only that job
ROM	Permanent memory, keeps data even without power	A printed book — you can read it, but not easily rewrite it
RAM	Temporary, fast "working" memory, cleared when power is off	A whiteboard — quick to write on and erase
DDR	A technique to send 2x the data per clock cycle	Getting two deliveries for the price of one trip

Self-Check Questions

1. What's the difference between a CPU and the ALU inside it?
2. Why is a GPU better than a CPU at rendering video, but worse at running a complex AI decision-making task?
3. A Raspberry Pi 5 with a Hailo-8 NPU attached will run AI models much faster than a plain Raspberry Pi 4B. Explain why, using the terms CPU, NPU, and RAM.
4. Why does data disappear from RAM when you turn off the power, but not from ROM?
5. In your own words, explain how DDR memory sends more data without increasing the clock speed.
6. Name the two types of RAM discussed that are used *inside* vs *around* the CPU, and explain the speed difference between them.